TS-06
CS-36

JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

KATOL ROAD, NAGPUR

An Autonomous Institute, with NAAC "A" Grade

Basic Science & Humanities Department

2023-24 (EVEN Semester)

CLASS TEST- I EXAMINATION

Course: B. Tech. **Subject:** Linear Algebra and Complex Analysis

Subject Code: CS/AI/DS 2T 001 **Year/Semester:** 2nd Semester (First Year)

Max Marks: 20 **Date :- 27/ 02/2024** **Duration:- 1 Hr**

Instructions to the Students:

1. All the questions carry marks as indicate.

2. Assume suitable data whenever necessary.

CO1.Describe concept of complex numbers, integral calculus & multiple integrals, probability and distribution theory.

CO2 Illustrate the concept of complex numbers, integral calculus & multiple integrals, probability and distribution theory by using examples.

CO3 Apply the knowledge of complex numbers, integral calculus & multiple integrals, probability and distribution theory to solve the engineering problems.

CO4 Analyze the problems and results of complex numbers, integral calculus & multiple integrals, probability and distribution theory to solve the real world problems.

CO5 Evaluate the problems by using complex numbers, integral calculus & multiple integrals, probability and distribution theory.

CO6 Create the methods or model by using complex numbers, integral calculus & multiple integrals, probability and distribution theory.

SECTION A

1. **All Questions are compulsory.**
2. **Answer in one word/sentence.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Define row matrix.	1/CO1	1
Q.2	Illustrate A + B if $A = \begin{bmatrix} 1 & 3 \\ 5 & 7 \end{bmatrix}$ and $B = \begin{bmatrix} -2 & 4 \\ -3 & 5 \end{bmatrix}$	2/CO2	1
Q.3	Define Scalar matrix.	1/CO1	1
Q.4	Define diagonal matrix.	1/CO1	1

SECTION B

1. **All Questions are compulsory.**
2. **Solve answer in brief (two, three Sentence)**

Q.5	Find Eigen value of matrix. $A = \begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$	1/CO1	2
Q.6	Illustrate Adjoint of $A = \begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$	2/CO2	2
Q.7	Find A. B where $A = \begin{bmatrix} 1 & -3 \\ 10 & 7 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ -1 & 5 \end{bmatrix}$	1/CO1	2
Q.8	Interpret the determinant of A, where $A = \begin{bmatrix} 2 & 3 \\ 5 & -7 \end{bmatrix}$	2/CO2	2

SECTION C

Solve any two questions.

Q.9	Identify Eigen values equation of matrix $A = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & -1 \\ 1 & -1 & 3 \end{bmatrix}$	3/CO3	4
Q.10	Identify the rank of $A = \begin{bmatrix} 1 & 2 & -4 & 5 \\ 2 & -1 & 3 & 6 \\ 8 & 1 & 9 & 7 \end{bmatrix}$	3/CO3	4
Q.11	Verify Cayley Hamilton theorem for $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$	4/CO4	4